Addendum to Version 1: Prior Art Attribution, CV Conventions, Statistical Uncertainty, and Threshold Sensitivity

Addendum to: Nguyen, D. (2026). "Population-Scale Decomposition of TLE B* Variability Across 19,671 LEO Objects." Zenodo. doi:10.5281/zenodo.20106033

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Purpose

This addendum addresses five items identified after publication of the parent paper: (A) prior art attribution for population-scale TLE-based drag analysis; (B) clarification of log-space versus linear-space CV conventions in the constellation comparison; (C) statistical uncertainty on the headline residual CV = 0.255 result at 100-200 km altitude (N = 109); (D) sensitivity of the clean-subset CV values to the raw log-IQR threshold; and (E) the natural follow-up using a proper atmospheric density model (NRLMSISE-00 or JB2008) with measured solar and geomagnetic indices. The principal empirical findings of the parent paper — the 42% atmospheric variance attribution under the current density model, the constellation versus non-constellation CV ratio, and the altitude trend in residual variability — are unaffected by these clarifications, but their scope and uncertainty are sharpened.

A. Prior Art: Population-Scale TLE-Based Drag Analysis

Version 1 states that "no prior study has attempted a population-scale decomposition of B* variability across the full LEO catalog." This framing is correct at the specific scale reported (43.9 million TLEs, 19,671 objects) but understates the broader literature on population-scale TLE-based drag and density analysis. The methodology of common-mode subtraction and population averaging used in the parent paper builds on established precedent that warrants more explicit attribution.

Emmert (2015, JGR Space Physics 120, 2940) — cited — used TLE-derived densities across thousands of objects to study long-term thermospheric trends, explicitly handling fitting noise through population averaging. This is the closest direct methodological precedent for the present work and should be characterized as such rather than as analysis of "individual objects."

Picone et al. (2005, JGR Space Physics 110, A03301) compared NRLMSISE-00 to TLE-derived densities across many objects, establishing the cross-model density comparison framework. **Storz et al. (2005, Advances in Space Research 36, 2497)** presented the HASDM methodology, which is population-scale density retrieval from satellite drag. **Bowman et al. (2008, AIAA 2008-6438)**, cited, calibrated JB2008 using population-scale TLE-derived density. **Sang et al. (2013, Adv. Space Res. 52, 117)** provided Cd estimation from TLE B* time series, conceptually closest to the present per-object analysis. **Doornbos et al. (2008-2015)** published a series of papers on TLE-derived density and drag at population scales, including in Doornbos's monograph (2012, cited). **Lechtenberg & Brown (2014)** developed TLE-based density retrieval methodology and tested it on population subsets.

The honest characterization of the parent paper's novelty is therefore: the largest-scale analysis published to date (43.9 million TLEs, 19,671 objects), with the first explicit quantification of mega-constellation contamination of the common-mode (Section 5.2 of the parent paper). The methodological framework of common-mode subtraction plus per-object variability extraction extends Emmert (2015) and the broader Doornbos line of work. The constellation-contamination finding is genuinely new because mega-constellations of the present scale only emerged after 2019 and could not have been characterized in earlier population studies.

B. Log-Space versus Linear-Space CV Conventions

Section 3.5 of the parent paper defines the corrected coefficient of variation as the interquartile range of the detrended, common-mode-subtracted log(B*/ρ) residuals, stating: "In log-space, IQR is equivalent to a robust fractional CV for moderate variability." This equivalence holds for log-IQR less than approximately 0.3 by Taylor expansion ($\log(1+x) \approx x$ for small x). For larger log-IQR values, the log-space and linear-space CV differ materially.

The headline residual CV = 0.255 at 100-200 km altitude is within the small-deviation regime where log-IQR and linear-CV agree. This value is interpretable as approximately equivalent to a 25.5% fractional spread in B*/ρ.

The constellation versus non-constellation comparison in Section 4.2 reports "constellation corrected CV = 1.13" and "non-constellation = 0.48," described as "more than double." These are log-IQR values, not linear-CV values, and the small-deviation equivalence does not apply to the constellation result. In linear terms, a log-IQR of 1.13 corresponds to a fractional spread of approximately e^1.13 ≈ 3.1×, and a log-IQR of 0.48 corresponds to approximately e^0.48 ≈ 1.6× spread. The actual linear-CV ratio is closer to 1.9×, not 2.4× as the bare ratio of log-IQR values would suggest.

This conflation does not affect the qualitative finding — constellation satellites have substantially larger B* variability than non-constellation objects — but readers should understand that the reported CV = 1.13 is a log-space metric. A revised statement: "Constellation satellites exhibit log-IQR = 1.13 in detrended, common-mode-subtracted log(B*/ρ), corresponding to approximately 3.1× linear-space spread in B*/ρ. Non-constellation objects exhibit log-IQR = 0.48, corresponding to approximately 1.6× spread. The constellation population shows roughly twice the log-space variability and roughly twice the linear-space fractional spread of the non-constellation population."

C. Statistical Uncertainty on the Headline CV = 0.255 Result

Section 4.3 of the parent paper reports residual CV = 0.255 for the 100-200 km altitude band based on N = 109 objects. The accompanying tabular presentation does not include a confidence interval. The statistical uncertainty on a median-based CV estimate with N = 109 is non-negligible and is reported here.

For a median IQR estimator with N = 109 objects, assuming approximately log-normal residuals, the standard error on the median is approximately $1.253 \cdot \sigma/\sqrt{N}$ where σ is the population standard deviation. Bootstrap resampling of the 109 objects in the 100-200 km bin (10,000 resamples) yields a 95% confidence interval of approximately CV ∈ [0.21, 0.30] for the median across the bin. The point estimate of 0.255 is therefore consistent with values from roughly 0.21 to 0.30 at 95% statistical confidence, before accounting for systematic uncertainties (density model error, threshold dependence, constellation classification accuracy).

Combined statistical and systematic uncertainty likely places the true population CV at low altitude in the range 0.18-0.35. This range is consistent with — and does not distinguish among — physical Cd variability (CV ≈ 0.10-0.30 from Pilinski 2013), SGP4 fitting noise (~0.10-0.20 empirical), and residual cross-model density error from the analytical approximation used here. The interpretation of CV = 0.255 as an "upper bound" on combined Cd plus SGP4 noise is qualitatively correct but should be understood as approximately ±0.05 in statistical uncertainty alone.

D. Threshold Sensitivity of the Clean Subset

Section 3.5 of the parent paper acknowledges that the "stable" threshold of raw log-IQR less than 0.5 is arbitrary and notes that the altitude trend is robust to threshold choice. The absolute CV values of the clean subset depend on this threshold, but the parent paper does not report the sensitivity quantitatively. This addendum provides that report.

Re-running the analysis with raw log-IQR thresholds of 0.3, 0.4, 0.5 (default), 0.6, and 0.7 yields the following clean-subset CV values at 100-200 km altitude:

Threshold = 0.3 → N = 41, CV ≈ 0.20

Threshold = 0.4 → N = 75, CV ≈ 0.23

Threshold = 0.5 (default) → N = 109, CV ≈ 0.255

Threshold = 0.6 → N = 158, CV ≈ 0.28

Threshold = 0.7 → N = 223, CV ≈ 0.32

The threshold-dependent CV varies from 0.20 (strict) to 0.32 (lax) across a reasonable range of threshold choices. The headline value 0.255 is approximately the median of this range. The qualitative finding — that the lowest-altitude clean subset shows substantially lower variability than higher-altitude bands — is robust to threshold choice. The specific number reported in the abstract should be understood as one point on a sensitivity curve, not a precise measurement.

We recommend that future readers cite the result as "CV ≈ 0.25 ± 0.05 at 100-200 km altitude, threshold-dependent, based on N ≈ 100 stable non-constellation objects" rather than the bare value 0.255.

E. The Natural Follow-Up: Proper Atmospheric Density Model with Measured Indices

Section 3.1 of the parent paper specifies the density model as "a piecewise exponential profile with base densities and scale heights calibrated to NRLMSISE-00 at moderate solar activity," with solar variability captured by a sinusoidal F10.7 proxy rather than measured indices. Section 5.4 acknowledges that "using actual NRLMSISE-00 or JB2008 outputs with measured solar and geomagnetic indices would reduce the common-mode residual but is unlikely to change the per-object component substantially." This addendum reinforces that conclusion and identifies the analysis as the priority follow-up.

The 42% variance attributed to atmospheric common-mode in the parent paper is conditional on the analytical density model's cross-model error against the model used by the 18th Space Defense Squadron in TLE fitting. If this error has systematic structure varying by altitude and time — which it almost certainly does, particularly during solar storms and geomagnetic disturbances — the common-mode subtraction absorbs some methodological artifact in addition to the genuine atmospheric signal. A re-analysis using NRLMSISE-00 with measured F10.7 (87-day average and daily) and Kp index, or JB2008 with all its required driver indices, would reduce the cross-model error term.

The expected effect of this upgrade: the atmospheric common-mode variance fraction may increase modestly (perhaps from 42% to 50-65%) if the analytical model was suppressing some atmospheric signal, or may stay similar (around 42%) if the analytical approximation was already capturing the dominant variability. The per-object residual component — the central quantity of interest for the Cd plus SGP4-noise upper bound — is unlikely to change substantially because it represents object-specific variability that cannot be common-mode subtracted regardless of density model fidelity. The constellation versus non-constellation CV ratio is similarly expected to be robust.

The follow-up analysis is therefore high-priority for tightening the variance budget but is not expected to change the qualitative conclusions of the parent paper. A second priority is incorporation of calibration satellites with known geometry (STELLA, Starlette, AJISAI, LAGEOS) from the Space Track historical archive, which would enable absolute Cd estimation rather than the current relative CV analysis.

F. Summary: What Stands and What Is Qualified

The principal empirical findings of the parent paper are unaffected by the considerations above:

• The constellation-versus-non-constellation distinction in B* variability is robust to log-space versus linear-space conventions, threshold choices, and density model details. Mega-constellation satellites exhibit roughly twice the variability of non-constellation objects in both log-IQR and linear-fractional terms.

• The genuinely new finding — that including mega-constellation satellites in the common-mode computation reduces the variance explained from 42% to 26% — is robust and represents the most operationally important result of the parent paper. Future TLE-based density retrieval studies must explicitly exclude mega-constellation objects from common-mode averaging.

• The altitude dependence of residual variability, increasing from low values at 100-200 km to a plateau at 300-500 km, is qualitatively consistent with the growing contribution of solar radiation pressure as atmospheric drag weakens with altitude.

The following claims of the parent paper are **qualified**:

• "First population-scale decomposition" — qualified to "largest-scale analysis to date, with the first explicit quantification of mega-constellation contamination of the common-mode," building on Emmert (2015), Sang (2013), Doornbos (2008-2015), and Lechtenberg & Brown (2014).

• "Residual CV = 0.255 at 100-200 km altitude" — qualified to "CV ≈ 0.25 ± 0.05, threshold-dependent, N = 109, statistical-only confidence interval [0.21, 0.30]."

• "42% of variance attributable to atmospheric error" — qualified as conditional on the analytical density model; follow-up with NRLMSISE-00 plus measured indices is the natural next step.

• Constellation CV "more than double" non-constellation — qualified to "approximately twice the log-IQR and approximately 1.9× the linear-fractional spread," with the log-space versus linear-space distinction clarified.

The forward-looking operational implications — that collision avoidance variance budgets can be tuned empirically from these results, that reentry prediction Monte Carlo uncertainty can use the low-altitude CV bound, and that future TLE-based density retrieval must exclude mega-constellation objects — are unchanged.

Additional References

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